2	**(a)**	**(i)**	base units of D:		
			force: $\text{kg}\,\text{m}\,\text{s}^{-2}$	B1	
			radius: m velocity: $\text{m}\,\text{s}^{-1}$	B1	
			base units of D: $[F / (R \times v)]$ $\text{kg}\,\text{m}\,\text{s}^{-2} / (\text{m} \times \text{m}\,\text{s}^{-1})$	M1	
			$= \text{kg}\,\text{m}^{-1}\,\text{s}^{-1}$	A0	[3]
		(ii)	**1.** $F = 6\pi \times D \times R \times v = [6\pi \times 6.6 \times 10^{-4} \times 1.5 \times 10^{-3} \times 3.7]$		
			$= 6.9 \times 10^{-5}$ N	A1	[1]
			2. $mg - F = ma$ hence $a = g - [F / m]$		
			$m = \rho \times V = \rho \times 4/3\ \pi\ R^3 = (1.4 \times 10^{-5})$	C1	
			$a = 9.81 - [6.9 \times 10^{-5}] / \rho \times 4/3\ \pi \times (1.5 \times 10^{-3})^3$ $(9.81 - 4.88)$	M1	
			$a = 4.9(3)\ \text{m}\,\text{s}^{-2}$	A1	[3]
	(b)	**(i)**	$a = g$ at time $t = 0$	B1	
			a decreases (as time increases)	B1	
			a goes to zero	B1	[3]
		(ii)	Correct shape below original line	M1	
			sketch goes to terminal velocity earlier	A1	[2]

3 (a) (i) work done equals force × distance moved / displacement in the direction of